Ketan Tiwari

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Portfolio
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Education

Shah & Anchor Kutchhi Engineering College Mumbai, India
Bachelor of Technology - Electronics and Computer Science; CGPA: 7.87 *2023-2027*

Skills Summary

- **Programming Languages** Python, Java, Javascript
- **Databases** PostgreSQL, NoSQL
- **Tools & Platforms** Git, GitHub, Visual Studio Code, Jupyter Notebook, Google Colab
- **AI & Machine Learning** Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), Generative AI (GenAI), Machine Learning, Deep Learning, Natural Language Processing (NLP)
- **Software Development** Backend Development, Frontend Development, REST APIs, Object-Oriented Programming (OOP), Data Structures & Algorithms (DSA), Software Development Life Cycle (SDLC)
- **Professional Skills** Problem Solving, Analytical Thinking, Communication, Team Collaboration

Experience

Backend Developer Intern | Sapphire Infocom Pvt. Ltd. | On-site **Sep 2025 – Dec 2025**

- Developed scalable backend systems using Node.js, Express.js and MongoDB, building RESTful APIs for efficient data processing and retrieval.
- Implemented JWT-based authentication to secure API endpoints and protect user data.

Generative AI Intern | Halliburton | On-site **Jun 2025 – Jul 2025**

- Developed an AI-powered Text-to-SQL chatbot using LLMs and RAG to convert natural language questions into Oracle SQL queries.
- Implemented schema-aware query generation on Halliburton's Oracle database, enabling accurate retrieval of enterprise well data.

Web Developer Intern | Oil and Natural Gas Corporation (ONGC) | On-site **Jul 2024 – Aug 2024**

- Designed and developed dynamic forms with input validation, backed by structured database storage and automated backend workflows to reduce manual entry errors.
- Built search and filtering mechanisms for efficient data querying and retrieval.

Projects

- **Natural Language Prompt to SQL Query Generator**
  - Developed an AI-powered Text-to-SQL chatbot using LLMs to convert natural language into SQL queries for enterprise databases.
  - Applied statistical analysis and Exploratory Data Analysis (EDA) on SQL query results to generate insights through interactive charts and visualizations.
  - Evaluated SQL query accuracy using F1-Score, Recall, MSE, and RMSE to measure model performance.
- **Multi-Agent AI Research System**
  - Built an autonomous multi-agent research pipeline using LangChain, Tavily, and BeautifulSoup for web search, content extraction, and automated report generation.
  - Implemented LCEL-based Writer and Critic agents for research synthesis, quality evaluation, and iterative refinement.
  - Developed a Streamlit UI to initiate research, track agent progress, and export structured research reports.
- **VISTA – AI Platform**
  - Designing and developing VISTA, an all-in-one AI platform integrating multiple AI capabilities including conversational AI, video intelligence, document analysis, image understanding, code assistance, data analysis, speech-to-text, and AI agents..
  - Built intelligent AI workflows using Python, LangChain, RAG, OpenAI, Mistral AI, Sarvam AI, and HuggingFace, enabling automated content understanding, reasoning, summarization, and structured output generation.
  - Implemented AI-powered agents and automation workflows that can perform multi-step tasks such as resume analysis, job requirement matching, opportunity search, comparison, and shortlisting through an interactive unified interface.
- **AI Interview Prep & Resume Analytics Platform**
  - Developed a Full-Stack AI Interview Platform using Node.js, Express, and React/SCSS, featuring a modular, responsive dashboard for mock interview workflows.
  - Integrated Generative AI APIs to dynamically analyze user inputs, generate tailored technical questions, and deliver real-time, context-aware performance feedback.
  - Engineered a Server-Side Resume Engine that programmatically parses user profiles and generates downloadable, professionally styled PDF documents.